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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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References/Text Books			
1	Introduction To Time Series Analysis And Forecasting, 2nd Edition, Wiley Series In Probability And Statistics, By Douglas C. Montgomery, Cheryl L. Jen(2015)		
2	Master Time Series Data Processing, Visualization, And Modeling Using Python Dr. Avishek Pal Dr. Pks Prakash (2017)		
3	Ruey S. Tsay "Analysis of Time-series data," Third Edition, Wiley 2014		

Faculty

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Academics

20/12/2025

UNIT-1
STATISTICAL MODELING & FORECASTING
LECTURE NO: -1

Basic Properties of time-series data: Distribution and moments, Stationarity, Autocorrelation, Heteroscedasticity, Normality:

1. **Distribution and Moments:** Time-series data may follow various distributions, such as normal, exponential, or Poisson. Moments like mean, variance, skewness, and kurtosis describe the shape and characteristics of the distribution.
2. **Stationarity:** A time series is stationary if its statistical properties like mean, variance, and autocorrelation structure do not change over time. This implies that the series has a constant mean and variance and the covariance between two time points only depends on the time lag between them.
3. **Autocorrelation:** Autocorrelation measures the degree of similarity between a time series and a lagged version of itself over different time lags. It helps in identifying patterns or dependencies within the data.
4. **Heteroscedasticity:** Heteroscedasticity refers to the situation where the variability of a variable is unequal across different values of another variable or across different time points in the case of time-series data. It violates the assumption of constant variance in many statistical models.
5. **Normality:** Normality refers to the assumption that the distribution of the residuals or errors in a statistical model follows a normal distribution. Deviations from normality can affect the validity of statistical inferences, especially in hypothesis testing and parameter estimation.

Basically there are **3 characteristics of a time series:** stationarity, trend and seasonality
Prerequisites: time series definition, statistics such as mean, variance, covariance.

An observed time series can be decomposed into three components: the trend (long term direction), the seasonal (systematic, calendar related movements) and the irregular (unsystematic, short term fluctuations)

One defining characteristic of a time series is that it is a list of observations where the ordering matters. Ordering is very important because there is dependency and changing the order could change the meaning of the data.

Basic concepts are the words that are necessary for comprehension of incoming information and performance of daily tasks. The correct understanding and usage of basic concepts is essential for effective communicative exchanges in your child's early years as well as success in academia in your child's later years.

The basic formula of Time Series Data:

$y(t) = x(t)\beta + \varepsilon(t)$, where $y(t) = \{y_t; t = 0, \pm 1, \pm 2, \dots\}$ is a sequence, indexed by the time subscript t , which is a combination of an observable signal sequence $x(t) = \{x_t\}$ and an unobservable white-noise sequence $\varepsilon(t) = \{\varepsilon_t\}$ of independently and identically distributed random variables.

We assume that the time series values we observe are the realizations of random variables $y_1, \dots, y_t$, which are in turn part of a larger stochastic process $\{y_t; t \in \mathbb{Z}\}$.

In time series analysis, the analogs to the mean and the variance are the mean function and the autocovariance function.

The mean of a series is defined as
 $\mu_t = E(y_t)$.

The autocovariance function is defined as
 $\gamma(s, t) = \text{cov}(y_s, y_t) = E[(y_s - \mu_s)(y_t - \mu_t)]$.

The autocovariance measures the linear dependence between two points (y_s, y_t) at different times. For smooth series the autocovariance function stays large even when the s and t are far apart, whereas for choppy series the autocovariance function is close to zero for large separations.

If $s=t$ it follows that

$$\gamma(t, t) = E[(y_t - \mu_t)^2] = \text{var}[y_t],$$

As in classical statistics, it is more convenient to deal with a measure of association between -1 and 1 . The autocorrelation function (ACF) is computed from the autocovariance function by dividing by the standard deviations of y_s and y_t .

$$\rho(s, t) = \gamma(s, t) / (\sigma_s \sigma_t)$$

The autocorrelation, also called serial correlation, is a measure of the internal correlation of a time series. It is a representation of the degree of similarity between the time series and a lagged version of itself. High autocorrelation values mean that the future is strongly correlated to the past.

The cross-covariance function is a measure of predictability of one series y_t from another series x_s .

$$\gamma_{xy}(s,t) = \text{cov}(x_s, y_t) = E[(x_s - \mu_{x_s})(y_t - \mu_{y_t})]$$

The cross-covariance function can be scaled to $[1, -1]$, referred to as cross-correlation function (CCF).

$$\rho_{xy}(s,t) = \gamma_{xy}(s,t) / (\gamma_x(s,s) \gamma_y(t,t))$$

Estimation of Correlation The real values for the mean and the autocorrelation function are in general not known and must be estimated based on the sample data $y_1, y_2, \dots, y_n$.

The mean function is estimated by the sample mean

$$\bar{y} = \frac{1}{n} \sum_{t=1}^n y_t \quad \text{and the theoretical autocorrelation function is estimated by the sample ACF}$$

$$\hat{\rho}^*(k) = \frac{\hat{\gamma}^*(k)}{\hat{\gamma}^*(0)} = \frac{\sum_{t=1}^{n-k} (y_{t+k} - \bar{y})(y_t - \bar{y})}{\sum_{t=1}^n (y_t - \bar{y})^2}, \text{ for } k=0, 1, \dots, n-1.$$

Stationarity and non-stationarity time series A key idea in time series analysis is that of stationarity. Stationarity is considered as an important precondition for the analysis of the correlational structure in the time series. A time series is considered stationary if its behavior does not change over time. This means, for example, that the values always tend to vary about the same level and that their variability is constant over time.

A time series $\{y_t: t \in \mathbb{Z}\}$ is said to be strictly stationary if for any $k > 0$ and any $t_1, \dots, t_k \in \mathbb{Z}$, the distribution of $(y_{t_1}, \dots, y_{t_k})$ is the same as that for $(y_{t_1+u}, \dots, y_{t_k+u})$ for every value of u . Following this definition the stochastic behavior of the process does not change through time.

A weaker definition of stationarity is second order stationarity, also referred to as wide-sense stationarity or covariance stationary. Here, we do not assume anything about the joint distribution of the random responses $y_{t_1}, y_{t_2}, y_{t_3}, \dots$ except that the mean is constant $E[y_t] = \mu$ and that the covariance between two observations y_t and y_{t+k} depends only on the lag k between two observations and not on the point t in the time series.

The theory for time series is based on the assumption of second-order stationarity. However, real-life data are often not stationary. The assumptions of stationarity above only applies after any trends/seasonal effects have been removed.

References/Text Books

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LECTURE NO: -2**Introduction to Time Series and Forecasting:**

- Time series analysis is a statistical method used to analyze and interpret data points collected sequentially over time.
- It involves identifying patterns, trends, and dependencies within the data to make predictions about future values.
- Forecasting, a key application of time series analysis, aims to predict future values based on historical data.
- Techniques such as moving averages, exponential smoothing, and ARIMA (Auto Regressive Integrated Moving Average) models are commonly used for time series forecasting.

These methods help businesses and researchers make informed decisions by providing insights into future trends and behaviors.

Definition and Features of a Time Series:

A time series is an ordered sequence of observations of the same phenomenon, which order is defined by the time when each data point is obtained. The data points are typically measured at equally spaced successive instants of time. Notice that time information is usually what accounts for the dependence observed in the data (the autocorrelation). The sample data are the observations x_{t1} , x_{t2} , ..., x_{tn} associated to times t_1 , t_2 , ..., t_n , respectively. As these observations are supposed to be dependent, some specific analytical tools are covered to adequately treat them. As aforementioned, the times are usually spaced at uniform time intervals, so the data might be denoted by x_t , $t = 1, 2, \dots, n$ (1.1) Sometimes, observed variables evolve and are registered continuously on time, so the notation will be x_t , $t_1 \leq t \leq t_2$ (1.2) In practice, however, discrete equally spaced times or periods are used, as expressed in 1.1. On the other hand, the two following situations are considered equivalent

- measuring a variable in successive instants of time, for example, observing the maximum ambient temperature each day
- measuring the total amount (accumulated value) of a variable of interest in a period defined by two consecutive time instants, for example, studying the total amount of tourists visiting a region each month.

Thus, clearly, a single observation will be available at each instant of time, which will condition the models and methodology chosen to study the time series. Such study aims to understand and describe the mechanism that generates the data, to later forecast future values and, sometimes, it is the starting point needed to establish a control mechanism of the observed system.

1.2 Classical

Decomposition Method. Examples. The first step in any analysis of a time series is the plot of the observations x_t say, against time t . Figure 1.1 shows the plots of some examples of time series with different behaviours. Traditionally the following model will be considered for the random variable observed: $X_t = T_t + S_t + v_t$ (1.3) It is called Additive Model and the components are:

- T_t , Trend. It represents the long term tendency of the series.
- S_t , Seasonality. It appears as oscillations around the trend, the cycles of which have constant period order s say. It compiles variations that can be observed periodically along the

evolution of the data. In natural and social sciences, the data are usually related with the seasons of the year so the period is a year. Some examples are the monthly crop, tourism behaviour, energy consumption, daily average temperature of a region or quarterly evolution of Gross Domestic Product of a country. In those cases, if the data is monthly, the periodicity order will be $s = 12$. Seasonal behaviour can also occur within the day, for example in the evolution of energy consumption, and in the week, in the evolution of the public transport use.

If the series has hourly data and the period of seasonality is the day, the periodicity order will be $s = 24$. If it has daily data and the period of seasonality is a week, the periodicity order will be $s = 7$.

- v_t , Random noise. Sometimes the following component will be added to the model (1.3)
- C_t , Cyclicity. Non-constant periodic oscillations. For example, in economics there are expansion and recession cycles. Cycles can also be found in some geophysics data series. If the cycles are long, it is difficult to differentiate them from the trend, so they can be considered together ($T_t + C_t$)

In some situation the Multiplicative Model might be more appropriate, that is: $X_t = T_t \cdot S_t \cdot v_t$. It can be transformed to additive via logarithms.

Exploring Time Series:

Data transformation in the plot of a time series is useful to observe trend and seasonality behaviour. These components contain the "macroscopic" evolution of the observations. Different methods are used to appreciate and describe the microscopic or random components. The expression can be rewritten as $X_t = T_t + S_t + v_t \approx \mu_t + v_t$ where μ_t is the evolution of the series after removing the random variations. Even though the concept will be introduced in detail in the next chapter, it is important to have an idea of the definition of stationarity to understand this last section. A series is stationary if the statistical properties such as mean, variance, autocovariance,... remain constant through time. In this situation, even if the values of the series themselves are not constant, there is a chance of being able to know what kind of behaviour to expect in the future. Stationary might be a reasonable assumption about what is left after trend and seasonality have been allowed for, so it is interesting to know methods to treat the data in order to find and/or eliminate these components. Least square minimization and data transformation are useful to find or eliminate some components of the series.

1.3.1 Least Square

minimization In some time series, data plot has a polynomial behaviour. For example USPOP can be estimated by an square polynomial and GNPSH by a line. In both situations, estimating the trend, will give an idea of the series behaviour. It is important to point out that the residuals will be not independent. 1.3.2 Data Transformation. Filtering Using a linear transformation of the series, it is possible to identify or eliminate a component, depending on the coefficients of the chosen linear transformation. To filter a time series X_t is to calculate a linear combination of the observations in order to obtain another time series Y_t with $Y_t = \sum_{j=-q}^q a_j X_{t+j}$ where q is a non-negative integer and a_j are constants. The filtering or smoothing effect is represented in the figure below. Usually the filter weights will be symmetric and normalized (That is $a_j = a_{-j}$ and $\sum_{j=-q}^q a_j = 1$)

Moving Average:

The moving average of a series construct a new series M_t obtained by the following expression: $M_t = \frac{1}{2q+1} \sum_{j=-q}^q X_{t+j}$ (1.4) The new time series $\{M_t\}$ is obtained from the original $\{X_t\}$ giving constant weights, $a_j = \frac{1}{2q+1}$, to $X_{t-q}, \dots, X_{t+q}$ and zero to the rest. This filter is a smoother of random noises. Check the effect of the command `filter(RANDOM, rep(1/15,15))` In a Series with seasonality order $s = 2q$ the following transformation will be used to eliminate the seasonality and identify the trend. $T_t = \frac{1}{s} \cdot (0.5 \cdot X_{t-q} + X_{t-q+1} + \dots + X_{t+q-1} + 0.5 \cdot X_{t+q})$.

Differencing:

First order differencing Operator, ∇ , is defined as $\nabla X_t = X_t - X_{t-1} = (1 - B)X_t$ where B is the backward shift operator, which is such as $B(X_t) = X_{t-1}$. ∇ is a linear filter because $\nabla X_t = X_t - X_{t-1} = \sum_{j=-1}^0 a_j X_{t+j}$

It is a linear combination of the observations with weights $a_0 = 1$, $a_{-1} = -1$ and $a_j = 0$ for $j \neq -1, 0$. The R command is `>diff`.

References/Text Books

- 1 " Introduction To Time Series Analysis And Forecasting, 2nd Edition, Wiley Series In Probability And Statistics, By Douglas C. Montgomery, Cheryl L. Jen(2015)".
- 2 "Master Time Series Data Processing, Visualization, And Modeling Using Python Dr. Avishek Pal Dr. Pks Prakash (2017)"
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LECTURE NO: -3**Different types of data:**

In time series analysis and forecasting, there are several types of data:

1. **Continuous Data:** Continuous data refers to measurements that can take any value within a given range. Examples include temperature, rainfall, stock prices, and GDP.
2. **Discrete Data:** Discrete data are countable and finite. They often represent events or occurrences that can be counted, such as the number of visitors to a website per hour or the number of products sold each day.
3. **Seasonal Data:** Seasonal data exhibit regular patterns or fluctuations that repeat over fixed periods, such as daily, weekly, monthly, or yearly. Examples include retail sales, energy consumption, and holiday bookings.
4. **Trend Data:** Trend data show a consistent upward or downward movement over time. Trends can be linear or nonlinear. Examples include population growth, technology adoption rates, and economic indicators.
5. **Cyclical Data:** Cyclical data exhibit fluctuations that are not of fixed period like seasonal patterns but occur due to economic or business cycles. Examples include real estate prices, unemployment rates, and business investment.
6. **Irregular Data:** Irregular data, also known as noise or random variation, do not follow a discernible pattern or trend. They represent unpredictable fluctuations in the time series data, often due to random events or measurement errors.
7. **Stationary Data:** Stationary data have constant mean, variance, and autocorrelation structure over time. They are easier to model and forecast because their statistical properties do not change over time.

References/Text Books

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LECTURE NO: -4**Internal structures of time series:**

Time series analysis and forecasting involve analyzing the internal structures of time series data to extract meaningful patterns and predict future values. Here are some common internal structures studied in time series analysis:

1. **Trend:** A long-term change in the series, either upwards or downwards. Trends can be linear or nonlinear.
2. **Seasonality:** Repeating patterns at regular intervals. For example, sales of ice cream might be higher in summer and lower in winter.
3. **Cycles:** Repeating up and down movements not necessarily of fixed duration. For example, business cycles of booms and busts.
4. **Irregularity:** Unpredictable movements that don't fit the trend or seasonality. These could be due to unforeseen events, random factors, or noise in the data.
5. **Autocorrelation:** The correlation of a time series with a delayed copy of itself. It measures the relationship between past observations and future ones.
6. **Stationarity:** A series is said to be stationary if its statistical properties, such as mean and variance, are constant over time. This simplifies analysis and modeling.
7. **Residuals:** The difference between observed and predicted values after fitting a model. Residual analysis helps assess the adequacy of a model.

To extract information from these structures and make forecasts, various techniques can be used:

These include:

1. **Smoothing:** Techniques like moving averages and exponential smoothing can help remove noise and reveal underlying trends or seasonality.
2. **Decomposition:** Breaking down a series into its trend, seasonality, and irregular components can aid in understanding its behavior.
3. **Autocorrelation Function (ACF):** This shows the correlation of a series with its past values at different lags. Significant peaks in the ACF can indicate seasonality or other patterns.
4. **Box-Jenkins Methodology:** Also known as ARIMA modeling, this approach combines Autoregressive (AR), Integrated (I), and Moving Average (MA) models to capture different aspects of a time series.
5. **Machine Learning:** Techniques such as Neural Networks, Support Vector

Machines, and RandomForests can also be used for forecasting, especially when dealing with non-linear data or large featurespaces.

Overall, understanding the internal structures of time series data is crucial for accurate analysis andforecasting. Various methods and techniques exist, each with its strengths and weaknesses, and the choice of approach depends on the specific characteristics of the data and the goals of the analysis.

References/Text Books

- 1 " Introduction To Time Series Analysis And Forecasting, 2nd Edition, Wiley Series In Probability And Statistics, By Douglas C. Montgomery, Cheryl L. Jen(2015)".
- 2 "Master Time Series Data Processing, Visualization, And Modeling Using Python Dr. Avishek Pal Dr. Pks Prakash (2017)"
- 3 Ruey S. Tsay "Analysis of Time-series data," Third Edition, Wiley 201

LECTURE NO: -5

Models for Time Series analysis:-

There are several models commonly used in time series analysis and forecasting:

- 1. Autoregressive Integrated Moving Average (ARIMA):** A popular model that combines autoregression, differencing, and moving average components to capture different aspects of the timeseries data.
- 2. Seasonal ARIMA (SARIMA):** Extends ARIMA to account for seasonal patterns in the data.
- 3. Exponential Smoothing Methods:** Includes models like Simple Exponential Smoothing, Holt's Exponential Smoothing, and Holt-Winters' Exponential Smoothing, which assign exponentially decreasing weights to past observations.
- 4. Vector Autoregression (VAR):** A model that generalizes autoregression to multivariate time series data, where each variable is regressed on its own lagged values as well as the lagged values of other variables in the system.
- 5. Long Short-Term Memory (LSTM):** A type of recurrent neural network (RNN) particularly suited for processing and predicting time series data, capable of capturing long-term dependencies.
- 6. Seasonal Decomposition of Time Series (STL):** A method for decomposing a time series into seasonal, trend, and residual components.
- 7. Gaussian Processes (GP):** A non-parametric Bayesian approach that models the distribution over functions directly, often used for time series regression and forecasting.
- 8. Prophet:** An open-source forecasting tool developed by Facebook, designed for forecasting timeseries data with multiple seasonality and holidays.

These are just a few examples, and the choice of model depends on various factors such as the characteristics of the data, the specific forecasting problem, computational resources, and the expertise available.

Models of time series analysis include: Classification: Identifies and assigns categories to the data. Curve fitting: Plots the data along a curve to study the relationships of variables within the data. Descriptive analysis: Identifies patterns in time series data, like trends, cycles, or seasonal variation.

Moving Average (MA), Autoregressive (AR), and Autoregressive Integrated Moving Average (ARIMA) are a few common examples. An ideal model must be selected based on each time series because different models will produce different results for the same set of data.

In time series quantitative data are arranged in the order of their occurrence and resulting statistical series. The quantitative values are usually recorded over equal time intervals such as daily, weekly, monthly, quarterly, half-yearly, yearly, or any other measure of time.

Some examples are statistics of Industrial Production in India on a monthly basis, birth-rate figures annually, the yield on ordinary shares, and weekly wholesale price of rice, etc. Components of Time Series There is a different kind of forces which influence the time series analysis. Some are continuously effective while others make themselves felt at recurring time intervals. So, our first task is to divide the data and elements into components.

A time series consists of the following four components or basic elements:

Basic or Secular or Long-time trend; Seasonal variations; Business cycles or cyclical movement; and Erratic or Irregular fluctuations. These components provide a basis for the explanation of the behavior on the past time. With their help, one can predict the behavior ahead. The major tendency of each component or constituent is largely due to causal factors.

Mathematical Statements of Time Series Some time series may not be affected by all type of variations. Some of these types of variations may affect a few time series only. Hence, while analyzing the time series, these effects are isolated. In a traditional time series analysis, we assume that any given observation consists of the trend, seasonal, cyclical and irregular movements.

Models of Time Series Analysis The following are the two models which we generally use for the decomposition of time series into its four components. The objective is to estimate and separate the four types of variations and to bring out the relative effect of each on the

overall behavior of the time series.

- (1) Additive model, and
- (2) Multiplicative model

1. Additive Model In the additive model, we represent a particular observation in a time series as the sum of these four components.

$$\text{i.e. } O = T + S + C + I$$

where O represents the original data, T represents the trend. S represents the seasonal variations, C represents the cyclical variations and I represents the irregular variations.

In another way, we can write $Y(t) = T(t) + S(t) + C(t) + I(t)$

2. Multiplicative Model In this model, four components have a multiplicative relationship. So, we represent a particular observation in a time series as the product of these four components:

$$\text{i.e. } O = T \times S \times C \times I$$

where O, T, S, C and I represents the terms as in additive model.

In another way, we can write $Y(t) = T(t) \times S(t) \times C(t) \times I(t)$

This model is the most used model in the decomposition of time series. To remove any doubt between the two models, it should be made clear that in Multiplicative model S, C, and I are indices expressed as decimal percentages whereas, in Additive model S, C and I are quantitative deviations about a trend that can be expressed as seasonal, cyclical and irregular in nature.

References/Text Books

- 1 " Introduction To Time Series Analysis And Forecasting, 2nd Edition, Wiley Series In Probability And Statistics, By Douglas C. Montgomery, Cheryl L. Jen(2015)".
- 2 "Master Time Series Data Processing, Visualization, And Modeling Using Python Dr. Avishek Pal Dr. Pks Prakash (2017)"
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LECTURE NO: -6**Examples of Time Series:**

Time series data can be found in various domains, and here are some examples:

1. **Stock Prices:** Daily, hourly, or even minute-by-minute prices of stocks or indices.
2. **Weather Data:** Temperature, humidity, precipitation, etc., recorded at regular intervals.
3. **Economic Indicators:** GDP, inflation rate, unemployment rate, etc., measured over time.
4. **Sales Data:** Daily, weekly, or monthly sales figures for products or services.
5. **Website Traffic:** Hourly or daily website visitors or page views.
6. **Energy Consumption:** Electricity usage measured at regular intervals.
7. **Healthcare Data:** Patient vital signs recorded over time, such as heart rate, blood pressure, etc.
8. **Social Media Activity:** Number of likes, shares, or comments on social media platforms over time.
9. **Population Statistics:** Census data tracking population growth or demographic changes over time.
10. **Sensor Data:** Readings from sensors in IoT devices, such as temperature sensors, motion sensors, etc.

Time series analysis: 6 usage examples:

It's always great to have solid examples of well-established businesses successfully using one or another technology, to see how it can potentially work in practice. Let's review six top time series analysis cases of companies across different business domains, highlighting an example of time series forecasting in business for each case:

1. Time Series in the Electronic Interconnect Industry: Case of a manufacturing company

The Intelliarts' customer is a service provider in the electronic interconnect industry, known for manufacturing and distributing connectors, cables, optics, and more.

Case description: The company faces challenges in predicting demand and improving stock availability. To address this, the company collaborated with the Intelliarts ML team to develop a machine learning prediction and forecasting engine. This engine is designed to detect patterns in the manufacturer's data and make accurate sales predictions, aiding in procurement forecasts.

Case outcomes: The implementation of this machine learning model has enabled Intelliarts' customer to anticipate future purchasing needs, manage resources more efficiently, and reduce costs and waste, leading to improved operational efficiency.

2. Time series in the financial sector: Case of Goldman Sachs

Goldman Sachs is a leading global investment banking, securities, and investment management firm. It's known for its expertise in market risk management.

Case description: Goldman Sachs employs time series analysis in market risk management, particularly for Value at Risk (VaR) modeling, which estimates potential portfolio losses. A real-life example of time series data could be the analysis of daily stock prices or market volatility. His approach enables Goldman Sachs to quantify the level of financial risk associated with their investment portfolios and make informed decisions to mitigate potential losses.

Case outcomes: The use of time series analysis enables Goldman Sachs to better manage market risk and make informed investment decisions. The firm's VaR modeling has been noted to average record levels, highlighting its significance in its risk management strategy.

3. Time series in the retail sector: Case of Walmart

Walmart is the world's largest retail corporation, offering an extensive range of products, including groceries, apparel, electronics, and home goods.

Case description: Walmart employs time series analysis to identify cycles on time series data in real business for demand forecasting and inventory management. By analyzing historical sales data, Walmart can anticipate future demand trends for various products and adjust its inventory accordingly.

Case outcomes: The application of time series analysis in Walmart's operations has led to significant improvements in supply chain efficiency and inventory management. By maintaining optimal stock levels and reducing the likelihood of stockouts, Walmart enhances customer satisfaction.

4. Time series in the energy sector: Case of National Grid

National Grid is an electricity and gas utility company responsible for energy transmission and distribution.

Case description: National Grid applies time series analysis for electricity demand forecasting, crucial for grid stability and energy distribution planning. By predicting the electricity demand, National Grid can ensure that there is a sufficient supply of energy to meet the needs of consumers and businesses.

Case outcomes: Enhanced grid stability and optimized energy distribution achieved through the application of time series analysis ensure reliable electricity supply preventing blackouts.

5. Time series in the technology sector: Case of Netflix

Netflix is a popular streaming service that offers a wide variety of TV shows, movies, and documentaries.

Case description: Netflix uses time series analysis to predict user engagement and content popularity, which are key factors in its content recommendation algorithm and strategic decision-making. By analyzing viewing patterns and trends, Netflix can tailor its recommendations to individual users.

Case outcomes: The data-driven approach has been useful in driving the company's subscriber growth and retention.

6. Time series in the telecommunications sector: Case of AT&T

AT&T is a telecommunications company providing a range of services, including wireless and wired internet, TV, and phone services.

Case description: AT&T employs time series analysis for network traffic forecasting, aiding in capacity planning and quality of service maintenance. By predicting the volume of data traffic on its network, AT&T can allocate resources efficiently, ensuring that its infrastructure can handle peak demand periods without compromising service quality.

Case outcomes: The proactive approach allows for enhanced network capacity planning and service quality, leading to improved customer satisfaction.

Interested in implementing a time series analysis solution in your business and enhancing it with AI? Drop Intelliarts a line, and let our expert data engineers contribute to your best project.

Final take:

Time series analysis is a powerful statistical tool used to forecast future trends, manage risks, optimize inventory, and enhance decision-making. By leveraging time series models, businesses can gain insights into patterns and seasonality in their data, leading to improved operational efficiency and competitive advantage. Implementing time series analysis, especially with AI enhancements, can significantly benefit companies in finance, retail, energy, technology, and telecommunications.

Implementing a time series solution in business, automating it, and enhancing it with AI capabilities, require substantial experience and expertise. The right call is to entrust your project to qualified engineers. Consider Intelliarts, a development firm with over 24 years of experience delivering cutting-edge and data-driven software solutions, your primary choice in this regard.

References/Text Books

- 1 " Introduction To Time Series Analysis And Forecasting, 2nd Edition, Wiley Series In Probability And Statistics, By Douglas C. Montgomery, Cheryl L. Jen(2015)".
- 2 "Master Time Series Data Processing, Visualization, And Modeling Using Python Dr. Avishek Pal Dr. Pks Prakash (2017)"
- 3 Ruey S. Tsay "Analysis of Time-series data," Third Edition, Wiley 2014

LECTURE NO: -7**Nature and uses of forecasting:-**

Time series analysis and forecasting involve examining historical data to make predictions about future values. Forecasting, in this context, serves several purposes:

1. **Predictive Modeling:** Forecasting allows us to model and predict future values based on historical patterns, enabling better decision-making in various fields such as finance, economics, and weather forecasting.
2. **Resource Allocation:** By forecasting future values, businesses can allocate resources more efficiently. For example, a retail store can forecast demand to optimize inventory levels and staffing.
3. **Planning and Strategy:** Forecasting helps in long-term planning and strategy development. For instance, a company can use sales forecasts to plan production schedules or expansion strategies.
4. **Risk Management:** Forecasting helps in identifying and mitigating risks. By predicting future trends and potential outcomes, organizations can better prepare for uncertainties and minimize their impact.
5. **Performance Evaluation:** Forecasts provide benchmarks against which actual outcomes can be compared. This helps in evaluating the effectiveness of strategies and making adjustments as needed.
6. **Policy Formulation:** Governments and policymakers use forecasts to formulate economic policies, allocate budgets, and address societal challenges such as unemployment and inflation.
7. **Research and Analysis:** Forecasting facilitates research and analysis by providing insights into underlying trends, patterns, and relationships within the data.

A **forecast** is a prediction of some future event or events. As suggested by Neils Bohr, making good predictions is not always easy. Famously “bad” forecasts include the following from the Predictions:

“The population is constant in size and will remain so right up to the end of mankind.”
L’Encyclopedie, 1756.

- “1930 will be a splendid employment year.” U.S. Department of Labor, New Year’s Forecast in 1929, just before the market crash on October 29.
- “Computers are multiplying at a rapid rate. By the turn of the century there will be 220,000 in the U.S.” Wall Street Journal, 1966.

Forecasting is an important problem that spans many fields including business and industry, government, economics, environmental sciences, medicine, social science, politics, and finance. Forecasting problems are often classified as short-term, medium-term, and long-term. Short-term forecasting problems involve predicting events only a few time periods (days, weeks, and months) into the future. Medium-term forecasts extend from 1 to 2 years into the future, and long-term forecasting problems can extend beyond that by many years. Short- and medium-term forecasts are required for activities that range from operations management to budgeting and selecting new research and development projects. Long-term forecasts impact issues such as strategic planning. Short- and medium-term forecasting is typically based on identifying, modeling, and extrapolating the patterns found in historical data. Because these historical data usually exhibit inertia and do not change dramatically very quickly, statistical methods are very useful for short- and medium-term forecasting. This book is about the use of these statistical methods.

Most forecasting problems involve the use of time series data. A **time series** is a time-oriented or chronological sequence of observations on a variable of interest. For example, Figure 1.1 shows the market yield on US Treasury Securities at 10-year constant maturity from April 1953 through December 2006 (data in Appendix B, Table B.1). This graph is called a **time**

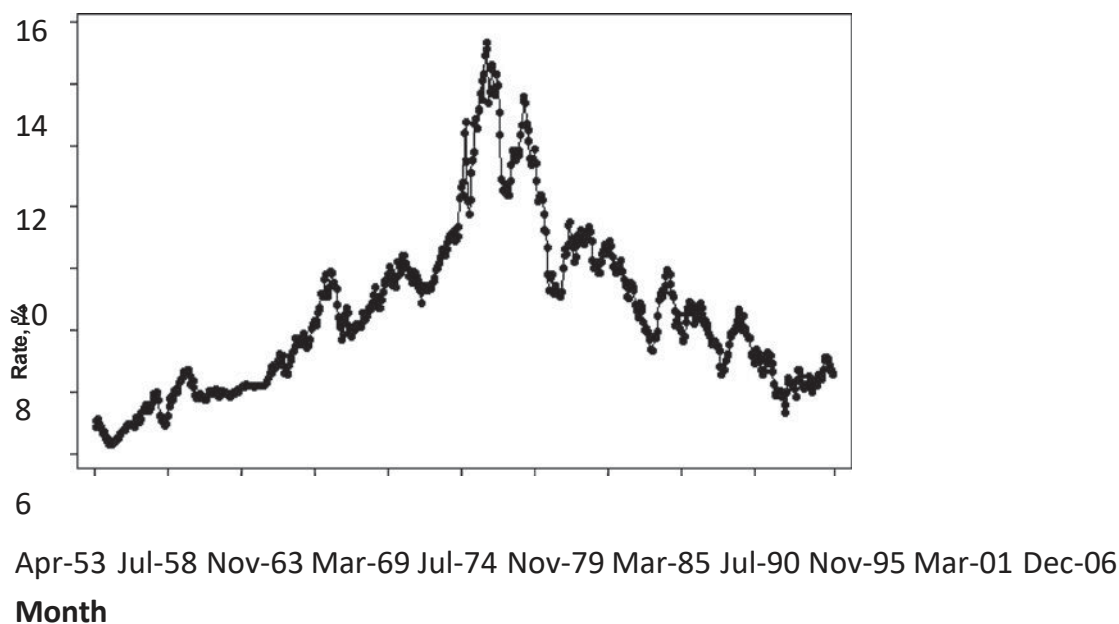


FIGURE 1.1 Time series plot of the market yield on US Treasury Securities at 10-year constant maturity. Source: US Treasury.

The rate variable is collected at equally spaced time periods, as is typical in most time series and forecasting applications. Many business applications of forecasting utilize daily, weekly, monthly, quarterly, or annual data, but any reporting interval may be used. Furthermore, the data may be instantaneous, such as the viscosity of a chemical product at the point in time where it is measured; it may be cumulative, such as the total sales of a product during the month; or it may be a statistic that in some way reflects the activity of the variable during the time period, such as the daily closing price of a specific stock on the New York Stock Exchange.

The reason that forecasting is so important is that prediction of future events is a critical input into many types of planning and decision-making processes, with application to areas such as the following:

- 1. Operations Management:** Business organizations routinely use forecasts of product sales or demand for services in order to schedule production, control inventories, manage the supply chain, determine staffing requirements, and plan capacity. Forecasts may also be used to determine the mix of products or services to be offered and the locations at which products are to be produced.
- 2. Marketing:** Forecasting is important in many marketing decisions. Forecasts of sales response to advertising expenditures, new promotions, or changes in pricing policies enable businesses to evaluate their effectiveness, determine whether goals are being met, and make adjustments.
- 3. Finance and Risk Management:** Investors in financial assets are interested in forecasting the returns from their investments. These assets include but are not limited to stocks, bonds, and commodities; other investment decisions can be made relative to forecasts of interest rates, options, and currency exchange rates. Financial risk management requires forecasts of the volatility of asset returns so that the risks associated with investment portfolios can be evaluated and insured, and so that financial derivatives can be properly priced.
- 4. Economics:** Governments, financial institutions, and policy organizations require forecasts of major economic variables, such as gross domestic product, population growth, unemployment, interest rates, inflation, job growth, production, and consumption. These forecasts are an integral part of the guidance behind monetary and fiscal policy, and budgeting plans and decisions made by governments. They are also instrumental in the strategic planning decisions made by business organizations and financial institutions.
- 5. Industrial Process Control:** Forecasts of the future values of critical quality characteristics of a production process can help determine when important controllable variables in the process should be changed, or if the process should be shut

down and overhauled. Feed- back and feedforward control schemes are widely used in monitoring and adjustment of industrial processes, and predictions of the process output are an integral part of these schemes.

6. Demography: Forecasts of population by country and regions are made routinely, often stratified by variables such as gender, age, and race. Demographers also forecast births, deaths, and migration patterns of populations. Governments use these forecasts for planning policy and social service actions, such as spending on health care, retirement programs, and antipoverty programs. Many businesses use forecasts of populations by age groups to make strategic plans regarding developing new product lines or the types of services that will be offered.

These are only a few of the many different situations where forecasts are required in order to make good decisions. Despite the wide range of problem situations that require forecasts, there are only two broad types of forecasting techniques—qualitative methods and quantitative methods.

Qualitative: forecasting techniques are often subjective in nature and require judgment on the part of experts. Qualitative forecasts are often used in situations where there is little or no historical data on which to base the forecast. An example would be the introduction of a new product, for which there is no relevant history. In this situation, the company might use the expert opinion of sales and marketing personnel to subjectively estimate product sales during the new product introduction phase of its life cycle. Sometimes qualitative forecasting methods make use of marketing tests, surveys of potential customers, and experience with the sales performance of other products (both their own and those of competitors). However, although some data analysis may be performed, the basis of the forecast is subjective judgment.

Perhaps the most formal and widely known qualitative forecasting technique is the **Delphi Method**: This technique was developed by the RAND Corporation (see Dalkey [1967]). It employs a panel of experts who are assumed to be knowledgeable about the problem. The panel members are physically separated to avoid their deliberations being impacted either by social pressures or by a single dominant individual. Each panel member responds to a questionnaire containing a series of questions and returns the information to a coordinator. Following the first questionnaire, subsequent questions are submitted to the panelists along with information about the opinions of the panel as a group. This allows panelists to review their predictions relative to the opinions of the entire group. After several rounds, it is hoped that the opinions of the panelists converge to a consensus, although achieving a consensus is not required and justified differences of opinion can be included in the outcome.

Quantitative forecasting techniques make formal use of historical data and a forecasting model. The model formally summarizes patterns in the data and expresses a statistical relationship between previous and current values of the variable. Then the model is used to project the patterns in the data into the future. In other words, the forecasting model is used to extrapolate past and current behavior into the future. There are several types of forecasting models in general use. The three most widely used are regression models, smoothing models, and general time series models. Regression models make use of relationships between the variable of interest and one or more related predictor variables. Sometimes regression models are called **causal forecasting models**, because the predictor variables are assumed to describe the forces that cause or drive the observed values of the variable of interest. An example would be using data on house purchases as a predictor variable to forecast furniture sales. The method of least squares is the formal basis of most regression models. **Smoothing models** typically employ a simple function of previous observations to provide a forecast of the variable of interest. These methods may have a formal statistical basis, but they are often used and justified heuristically on the basis that they are easy to use and produce satisfactory results. General **time series models** employ the statistical properties of the historical data to specify a formal model and then estimate the unknown parameters of this model (usually) by least squares. In subsequent chapters, we will discuss all three types of quantitative forecasting models.

The form of the forecast can be important. We typically think of a forecast as a single number that represents our best estimate of the future value of the variable of interest. Statisticians would call this a **point estimate** or **point forecast**. Now these forecasts are almost always wrong; that is, we experience **forecast error**. Consequently, it is usually a good practice to accompany a forecast with an estimate of how large a forecast error might be experienced. One way to do this is to provide a **prediction interval (PI)** to accompany the point forecast. The PI is a range of values for the future observation, and it is likely to prove far more useful in decision-making than a single number.

Other important features of the forecasting problem are the **forecast horizon** and the **forecast interval**. The forecast horizon is the number of future periods for which forecasts must be produced. The horizon is often dictated by the nature of the problem. For example, in production planning, forecasts of product demand may be made on a monthly basis. Because of the time required to change or modify a production schedule, ensure that sufficient raw material and component parts are available from the supply chain, and plan the delivery of completed goods to customers or inventory facilities, it would be necessary to forecast up to 3 months ahead. The forecast horizon is also often called the forecast lead time. The forecast interval is the frequency with which new forecasts are prepared. For example, in production planning, we might forecast demand on a monthly basis, for up to 3 months in the future (the lead time or horizon), and

prepare a new forecast each month. Thus the forecast interval is 1 month, the same as the basic period of time for which each forecast is made. If the forecast lead time is always the same length, say, T periods, and the forecast is revised each time period, then we are employing a rolling or moving horizon forecasting approach. This system updates or revises the forecasts for $T-1$ of the periods in the horizon and computes a forecast for the newest period T . This rolling horizon approach to forecasting is widely used when the lead time is several periods long.

References/Text Books

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LECTURE NO: -8

Forecasting Process:

The forecasting process in time series:

It is usually chosen as some multiple of the smallest seasonal cycle in the data. So, if the data are monthly and exhibit a monthly cycle, k would be a multiple.

Regression imputation is a variation of mean value imputation where the imputed value is computed from a model used to predict the missing value. The prediction model does not have to be a linear regression model. For example, it could be a time series model.

Hot deck imputation is an old technique that is also known as the last value carried forward method. The term “hot deck” comes from the use of computer punch cards. The deck of cards was “hot” because it was currently in use. **Cold deck imputation** uses information from a deck of cards not currently in use. In hot deck imputation, the missing values are imputed by using values from similar complete observations. If there are several variables, sort the data by the variables that are most related to the missing observation and then, starting at the top, replace the missing values with the value of the immediately preceding variable. There are many variants of this procedure.

What is Forecasting?

Forecasting is about making smart guesses about what might happen in the future by looking at past information and patterns using math-based methods. It's important for making choices, planning, and dealing with risks in areas, like business, money matters, economics, and even weather forecasts. There are two main types of forecasting: qualitative forecasting and quantitative forecasting.

Qualitative forecasting uses the ideas and thoughts of experts to guess what might happen in the future. It is helpful when there is not much past information or when it is hard to measure the factors affecting the situation. Quantitative forecasting uses math and numbers to look at past data and find patterns or trends. It works best when there's a lot of reliable past information and when the connections between different things can be measured with numbers. We will talk about the types in detail under the head techniques.

Steps in the Process of Forecasting:

The process of forecasting involves a series of steps to make sure that predictions are accurate and help in effective decision-making. Here are the key steps in the forecasting process, explained in a simple and easy-to-understand way:

- **Set clear goals:** Start by deciding what you want to predict and why. Figure out the purpose of the forecast, and choose the time frame you're interested in, whether short-term, medium-term, or long-term.
- **Gather information:** Collect past data related to your forecast, like previous sales numbers, market changes, or economic signs. Make sure the information is correct, trustworthy, and up-to-date to get the best possible forecast.
- **Choose a method:** Pick the right forecasting technique based on your goals and the available data. There are various methods, including qualitative approaches (like expert opinions) and quantitative methods (such as statistical models and algorithms).
- **Analyse the data:** Use the chosen method to examine the data and look for patterns, trends, or relationships. This analysis will help you make informed predictions about future events or outcomes.
- **Make the forecast:** Based on your analysis, make educated guesses about what might happen in the future. Keep in mind the limitations of forecasting and the uncertainty of the future while making your predictions.
- **Validate the forecast:** Check your predictions against actual outcomes or historical data to see how accurate your forecast is. This step helps identify any issues or inaccuracies in the forecasting process and can guide improvements in future forecasts.
- **Review and adjust:** Regularly review your forecast and update it as new information becomes available or conditions change. Stay flexible and be prepared to modify your predictions and plans as needed.

Techniques of Forecasting:

Forecasting techniques are important tools for businesses and managers to make informed decisions about the future. By using these techniques, they can anticipate future trends and make plans to succeed in the long term. Some of the techniques are explained below :

Time Series Analysis: It is a method of analyzing data that is ordered and time-dependent, commonly used in fields such as finance, economics, engineering, and social sciences. This method involves decomposing a historical series of data into various components, including trends, seasonal variations, cyclical variations, and random variations. By separating the various components of a time series, we can

identify underlying patterns and trends in the data, and make predictions about future values. The trend component represents the long-term movement in the data, while the seasonal component represents regular, repeating patterns that occur within a fixed time interval. The cyclical component represents longer-term, irregular patterns that are not tied to a fixed time interval, and the random component represents the unpredictable, random fluctuations that are present in any time series.

Extrapolation: It is a statistical method used to estimate values of a variable beyond the range of available data by extending or projecting the trend observed in the existing data. It is commonly used in fields, such as economics, finance, engineering, and social sciences to predict future trends and patterns. To perform extrapolation, various methods can be used, including linear regression, exponential smoothing, and time series analysis. The choice of method depends on the nature of the data and the type of trend observed in the existing data.

Regression Analysis: Regression analysis is a statistical method used to analyze the relationship between one or more independent variables and a dependent variable. The dependent variable is the variable that we want to predict or explain, while the independent variables are the variables that we use to make the prediction or explanation. It can be used to identify and quantify the strength of the relationship between the dependent variable and independent variables, as well as to make predictions about future values of the dependent variable based on the values of the independent variables.

Input-Output Analysis: Input-Output Analysis is a method of analyzing the interdependence between different sectors of an economy by examining the flows of goods and services between them. This method helps to measure the economic impact of changes in production, consumption, and investment in a given economy. The fundamental principle of Input-Output Analysis is that each sector of an economy depends on other sectors for the supply of goods and services, and also provides goods and services to other sectors. These interdependencies create a network of transactions between sectors, which can be represented using an input-output table.

Historical Analogy: Historical analogy is a method of reasoning that involves comparing events or situations from the past with those in the present or future. This method is used to gain insights into current events or to make predictions about future events by looking at similar events or situations in the past. The premise of historical analogy is that history repeats itself, and that by studying past events, we can gain an understanding of the factors that led to those events and how they might play out in similar situations. For instance, political analysts may use the analogy of the rise of fascism in Europe in the 1930s to understand the current political climate in a particular country.

Business Barometers: Business barometers are statistical tools which are used to measure and evaluate the overall health and performance of a business or industry. These barometers are based on various economic indicators, such as sales figures, production data, employment rates, and consumer spending patterns. The main purpose of a business barometer is to provide an objective and quantitative measure of the current and future state of a business or industry. By analyzing these economic indicators, business owners and managers can make informed decisions about their operations and strategies.

Panel Consensus Method: The Panel Consensus Method is a decision-making technique that involves a group of experts sharing their opinions and experiences on a particular topic. The goal of this method is to arrive at a consensus or agreement among the group on the best course of action. In the Panel Consensus Method, a panel of experts is selected based on their knowledge and experience in the relevant field. The panel is presented with a problem or issue to be addressed, and each member provides their opinion or recommendation. The panel members then discuss their opinions and try to reach a consensus on the best course of action. It can be used in various fields, such as healthcare, business, and public policy, among others. It is particularly useful in situations where there is no clear-cut solution to a problem, and multiple viewpoints need to be considered.

Delphi Technique: The Delphi Technique is a decision-making process that involves a group of experts providing their opinions and insights on a particular topic or problem. This method is designed to reach a consensus on a course of action using a structured and iterative approach. In this, a facilitator presents a problem or question to a group of experts, who then provide their opinions or recommendations. The facilitator collects the responses and presents them to the group anonymously. The experts review the responses and provide feedback, revisions, or additions to the responses. This process is repeated until a consensus is reached.

Morphological Analysis: Morphological Analysis is a problem-solving method that involves breaking down a complex problem or system into smaller components, referred to as "morphological variables". These variables are then analyzed to identify potential solutions or courses of action. It begins by assembling a team of experts or stakeholders to identify the variables contributing to the problem or system. These variables may be identified through brainstorming or other techniques

References/Text Books

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Data for forecasting:

The appropriate forecasting methods depend largely on what data are available.

If there are no data available, or if the data available are not relevant to the forecasts, then qualitative forecasting methods must be used. These methods are not purely guesswork—there are well-developed structured approaches to obtaining good forecasts without using historical data. Quantitative forecasting can be applied when two conditions are satisfied:

1. numerical information about the past is available;
2. it is reasonable to assume that some aspects of the past patterns will continue into the future.

There is a wide range of quantitative forecasting methods, often developed within specific disciplines for specific purposes. Each method has its own properties, accuracies, and costs that must be considered when choosing a specific method.

Most quantitative prediction problems use either time series data (collected at regular intervals over time) or cross-sectional data (collected at a single point in time). In this book we are concerned with forecasting future data, and we concentrate on the time series domain.

Examples of time series data include:

- Daily IBM stock prices
- Monthly rainfall
- Quarterly sales results for Amazon
- Annual Google profits

Predictor variables and time series forecasting: Predictor variables are often useful in time series forecasting. For example, suppose we wish to forecast the hourly electricity demand (ED) of a hot region during the summer period. A model with predictor variables might be of the form $ED_t = f(\text{current temperature, strength of economy, population, time of day, day of week, error})$.

The relationship is not exact — there will always be changes in electricity demand that cannot be accounted for by the predictor variables. The “error” term on the right allows for random variation and the effects of relevant variables that are not included in the model. We call this an explanatory model because it helps explain what causes the variation in electricity demand.

Because the electricity demand data form a time series, we could also use a time series

UNIT 4: Time Series Forecasting. In this case, a simple time series forecasting model is of the

form ED_{t+1}

$f(ED_t, ED_{t-1}, ED_{t-2}, ED_{t-3}, \dots, \text{error})$,

where

t is the present hour,

$t+1$ is the next hour,

$t-1$ is the previous hour,

$t-2$ is two hours ago, and so on. Here, prediction of the future is based on past values of a variable, but not on external variables which may affect the system. Again, the "error" term on the right allows for random variation and the effects of relevant variables that are not included in the model.

There is also a third type of model which combines the features of the above two models.

For example, it might be given by ED_{t+1}

$f(ED_t, \text{current temperature, time of day, day of week, error})$.

These types of mixed models have been given various names in different disciplines. They are known as dynamic regression models, panel data models, longitudinal models, transfer function models, and linear system models (assuming that f is linear). These models are discussed in Chapter 9.

An explanatory model is useful because it incorporates information about other variables, rather than only historical values of the variable to be forecast. However, there are several reasons a forecaster might select a time series model rather than an explanatory or mixed model. First, the system may not be understood, and even if it was understood it may be extremely difficult to measure the relationships that are assumed to govern its behaviour. Second, it is necessary to know or forecast the future values of the various predictors in order to be able to forecast the variable of interest, and this may be too difficult. Third, the main concern may be only to predict what will happen, not to know why it happens. Finally, the time series model may give more accurate forecasts than an explanatory or mixed model.

References/Text Books

1 " Introduction To Time Series Analysis And Forecasting, 2nd Edition, Wiley Series In Probability And Statistics, By Douglas C. Montgomery, Cheryl L. Jen(2015)".

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Resources for forecasting:-

There are several resources you can explore to learn about time series analysis and forecasting:

1. Books:

- "Time Series Analysis and Its Applications: With Examples" by Robert H. Shumway and David S. Stoffer.
- "Forecasting: Principles and Practice" by Rob J. Hyndman and George Athanasopoulos. This book is available online for free.
- "Introduction to Time Series and Forecasting" by Peter J. Brockwell and Richard A. Davis.

2. Online Courses:

- Course offers courses like "Time Series Analysis in Python" and "Forecasting Models for Marketing Decisions."
- Udemy has courses such as "Time Series Analysis and Forecasting using Python" and "Practical Time Series Data Analysis and Visualization."

3. Websites and Blogs:

- Towards Data Science: A platform on Medium with numerous articles on time series analysis and forecasting.
- Kaggle: Explore datasets and participate in competitions related to time series forecasting. You can also find kernels (code notebooks) shared by others.

4. Software Documentation:

- Documentation for programming languages and libraries such as Python's pandas, statsmodels and scikit-learn. Also, documentation for R packages like forecast, tsibble and TTR (Technical Trading Rules).

5. Research Papers and Journals:



- Journals like the Journal of Time Series Analysis and the International Journal of Forecasting publish research articles on time series analysis and forecasting.

6. Online Communities:

- Join communities like Stack Overflow, Cross Validated (part of Stack Exchange), and Reddit's r/datascience to ask questions and learn from others' experiences.

7. YouTube Channels:

Channels like "StatQuest with Josh Starmer" and "Data School" offer tutorials on time series analysis and forecasting.

There are a variety of good resources that can be helpful to technical professionals involved in developing forecasting models and preparing forecasts. There are three professional journals devoted to forecasting:

- Journal of Forecasting
- International Journal of Forecasting
- Journal of Business Forecasting Methods and Systems

We recommend Box, Jenkins, and Reinsel (1994); Chatfield (1996); Fuller (1995); Abraham and Ledolter (1983); Montgomery, Johnson, and Gardiner (1990); Wei (2006); and Brockwell and Davis (1991, 2002). Some of these books are more specialized than this one, in that they focus on a specific type of forecasting model such as the autoregressive integrated moving average [ARIMA] model, and some also require more background in statistics and mathematics.

Many statistics software packages have very good capability for fitting a variety of forecasting models. Minitab Statistical Software, JMP®, the Statistical Analysis System (SAS) and R are the packages that we utilize and illustrate in this book. At the end of most chapters we provide R code for working some of the examples in the chapter. Matlab and S-Plus are also two packages that have excellent capability for solving forecasting problems.

References/Text Books

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